



CODE

Lorem ipsum, or lipsum as it is sometimes Lorem



PLAN

Lorem ipsum, or lipsum as it is sometimes Lorem



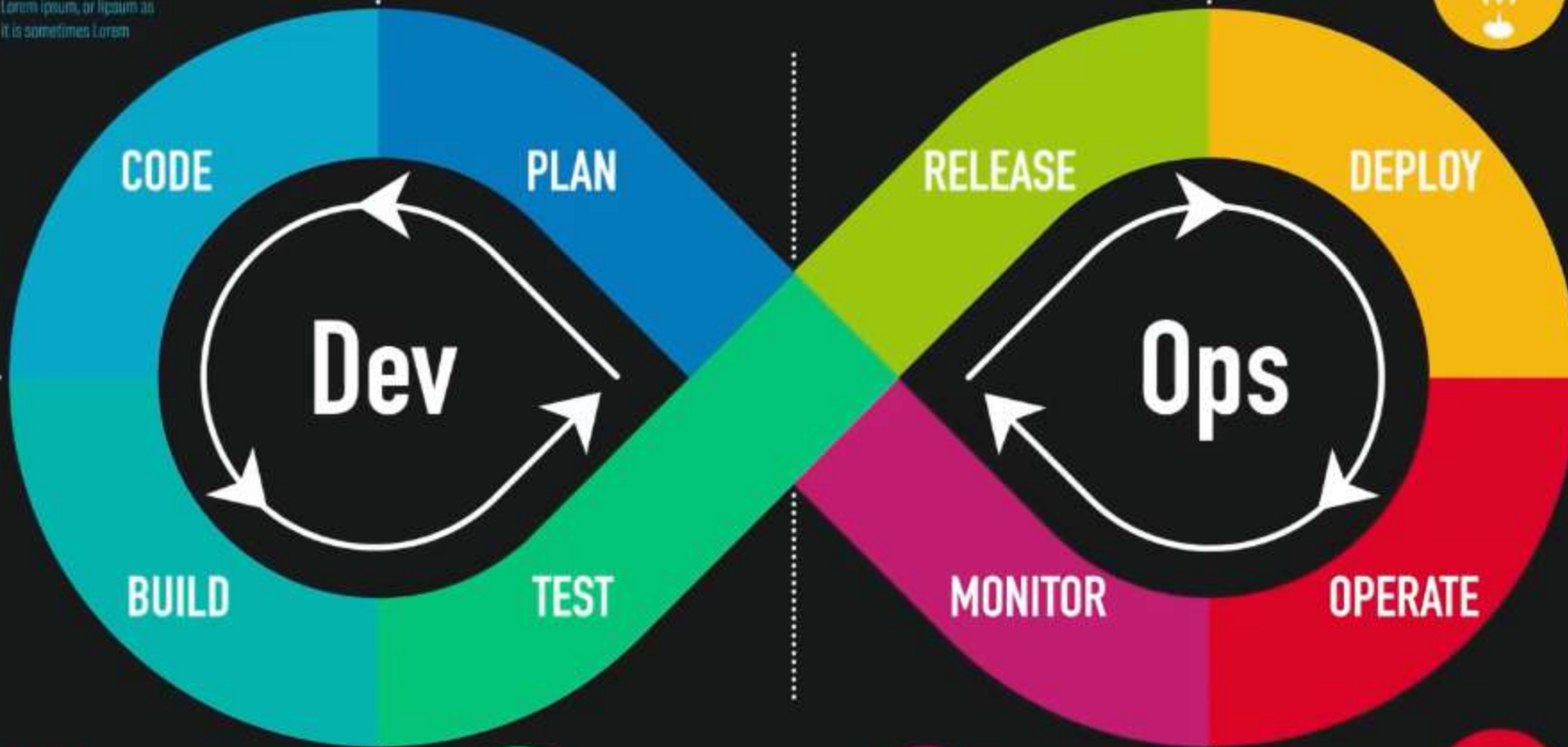
RELEASE

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DEPLOY

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BUILD

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TEST

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MONITOR

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OPERATE

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DEVOPS



ROAD MAP

LINUX FUNDAMENTALS



LINUX FUNDAMENTALS



BASH
THE BOURNE-AGAIN SHELL



3 WEEKS

LINUX FUNDAMENTALS

- ✓ **File system: ls, cp, mv, rm**
- ✓ **Permissions: chmod, chown**
- ✓ **Processes: ps, top, kill**
- ✓ **Packages: apt, yum**

NETWORKING CONCEPTS





NETWORKING CONCEPTS



WIRESHARK



2 WEEKS



NETWORKING CONCEPTS

- ✓ **OSI and TCP/IP models**
- ✓ **IP addressing and subnetting**
- ✓ **DNS and DHCP**
- ✓ **HTTP, HTTPS, FTP, and SSH**
- ✓ **Firewalls and security groups**
- ✓ **Ping, traceroute, and netstat**



GIT



2 WEEKS



GIT

- ✓ **Clone, commit, push, and pull**
- ✓ **Branching and merging**
- ✓ **Resolving merge conflicts**
- ✓ **Working with remote repositories**

PROGRAMMING LANGUAGES



6 WEEKS

PROGRAMMING LANGUAGES

- ✓ Python syntax & data structures
- ✓ Modules & packages
- ✓ Write and execute Python scripts
- ✓ Work with files
- ✓ Handle errors
- ✓ Write automation scripts



CLOUD PROVIDERS



6 WEEKS



CLOUD PROVIDERS

- ✓ **Launch, configure, & manage virtual servers**
- ✓ **Store & manage data**
- ✓ **Manage users, groups, and roles**
- ✓ **Setup and manage isolated networks**

CONTAINERIZATION



4 WEEKS

Container 1

App 1

Libraries

Container 2

App 2

Libraries

Container 3

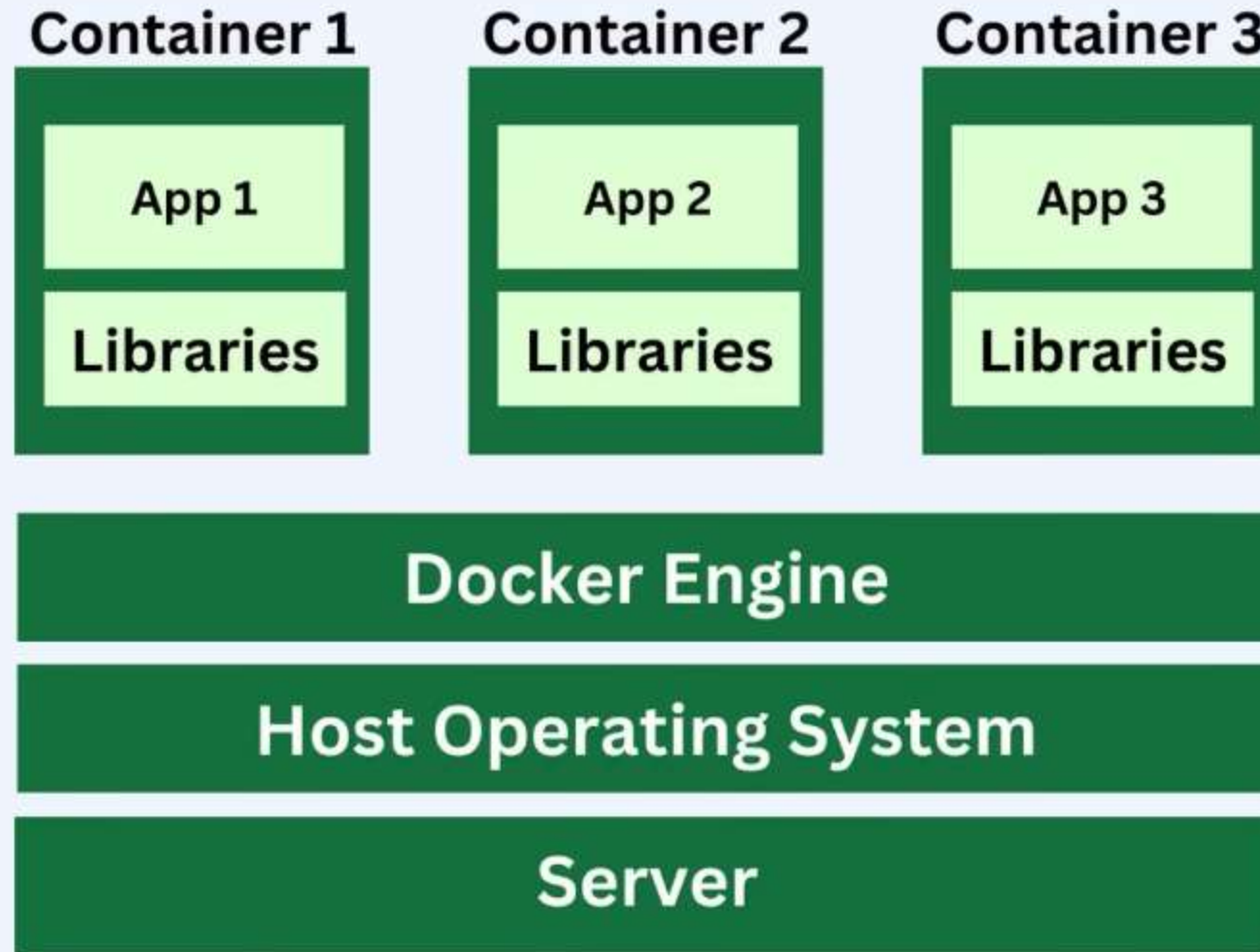
App 3

Libraries

Docker Engine

Host Operating System

Server



CONTAINERIZATION

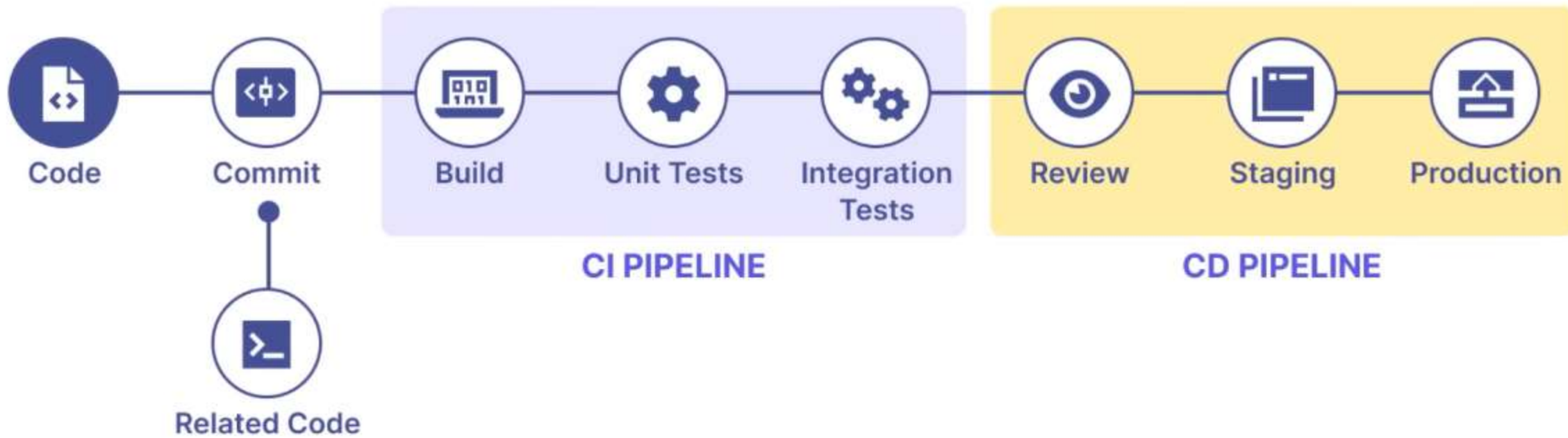
- ✓ Create Docker images
- ✓ Starting, stopping, & managing containers
- ✓ Write Dockerfiles
- ✓ Run multi-container apps using Docker Compose

- • **CONTINUOUS INTEGRATION/DEPLOYMENT**

CI/CD



4 WEEKS



• • CONTINUOUS INTEGRATION/DEPLOYMENT

CI/CD

- ✓ Create and manage Jenkins pipelines
- ✓ Write Jenkinsfiles
- ✓ Integrate automated tests
- ✓ Automate the build process
- ✓ Automate the deployment

ORCHESTRATION & MANAGEMENT



6 WEEKS

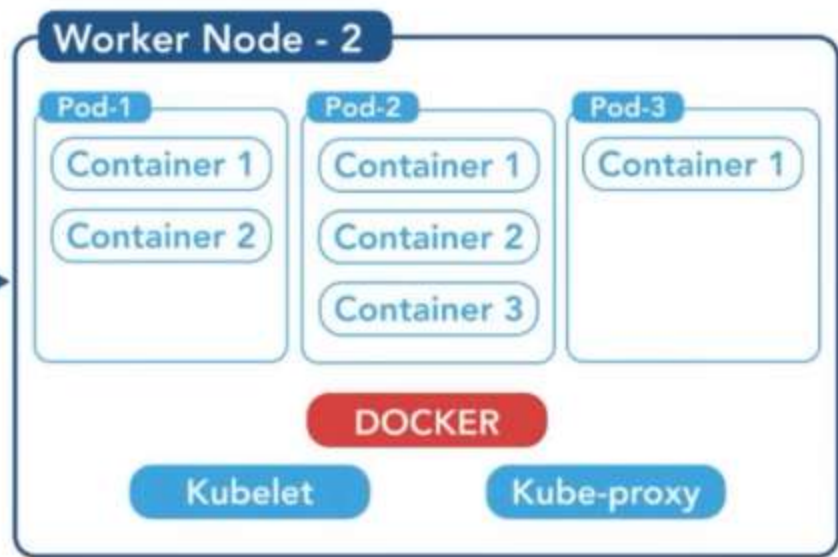
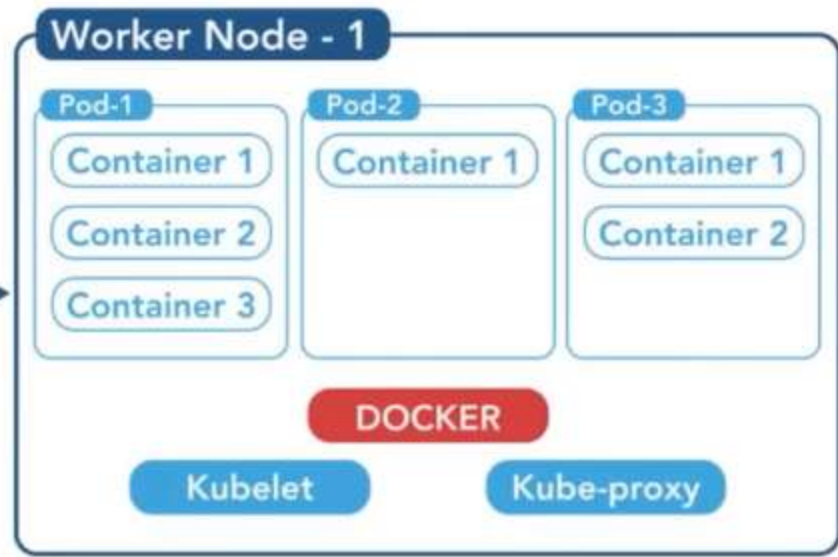
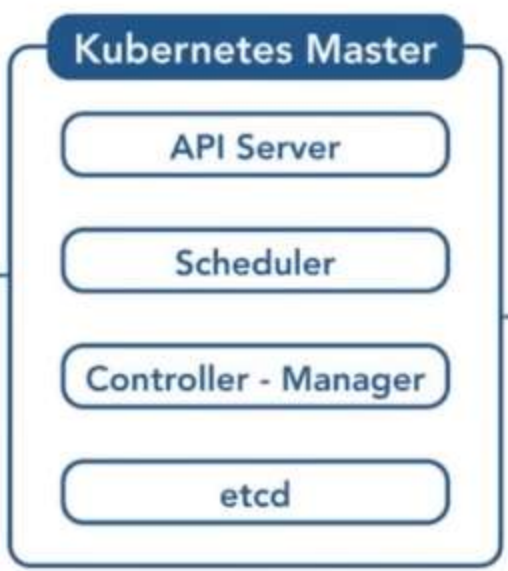
Kubernetes Architecture



User Interface



Kubectl



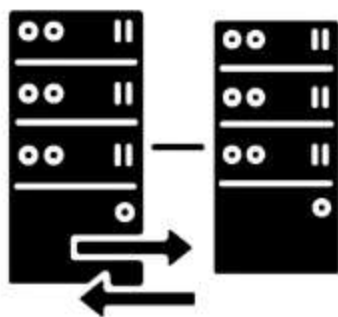
• • • ORCHESTRATION & MANAGEMENT

- ✓ Overall architecture
- ✓ Key components
- ✓ Manage resources
- ✓ Scale applications
- ✓ Networking model

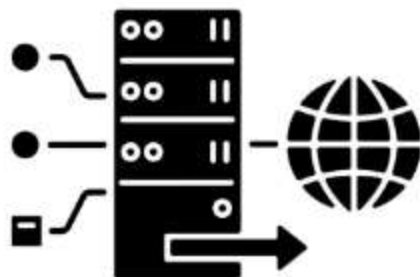
• • NETWORKING & INFRASTRUCTURE SERVICES



4 WEEKS



**REVERSE
PROXIES**



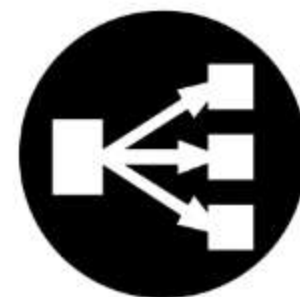
**FORWARD
PROXIES**



**CACHING
SERVERS**



FIREWALLS



**LOAD
BALANCERS**

NETWORKING & INFRASTRUCTURE SERVICES

- ✓ **Configure Nginx as a reverse proxy**
- ✓ **Configure it to act as a forward proxy**
- ✓ **Caching strategies**
- ✓ **Configure firewalls & security groups**

... CONFIGURATION MANAGEMENT



4 WEEKS

• • • **CONFIGURATION MANAGEMENT**

- ✓ **Write Ansible playbooks**
- ✓ **Use roles & modules**
- ✓ **Manage variables & templates**

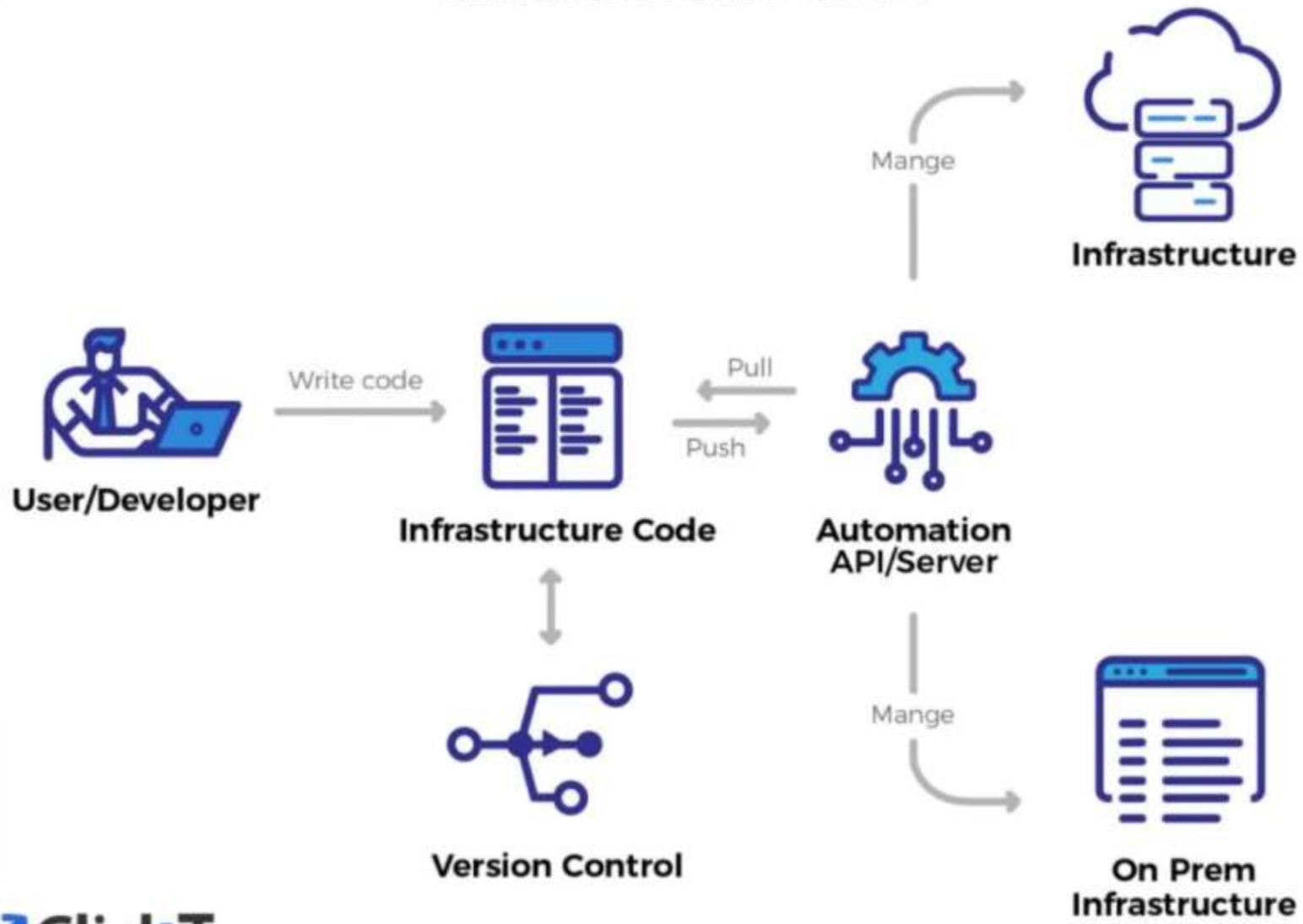
..... INFRASTRUCTURE AS CODE

IaC



4 WEEKS

Infrastructure Code



..... **INFRASTRUCTURE AS CODE**

IaC

- ✓ **Basic concepts**
- ✓ **Terraform configuration files**
- ✓ **Terraform modules**
- ✓ **Advanced concepts**

MONITORING & LOGGING



Prometheus



Grafana



4 WEEKS



MONITORING & LOGGING

- ✓ **Architecture and data model**
- ✓ **Collect metrics**
- ✓ **Write queries**
- ✓ **Set up alerts**



10-14 MONTHS TOTAL

Skill	Est. Time	Learning Phase
Linux Fundamentals	2-3 weeks	Beginner
Networking Concepts	2 weeks	Beginner
Version Control (Git)	1-2 weeks	Beginner
Programming Languages (Python)	4-6 weeks	Beginner
Cloud Providers (AWS)	4-6 weeks	Intermediate
Containerization (Docker)	3-4 weeks	Intermediate
CI/CD (Jenkins)	3-4 weeks	Intermediate
Container Orchestration (Kubernetes)	4-6 weeks	Intermediate
Networking Services (Nginx)	3-4 weeks	Intermediate
Configuration Management (Ansible)	3-4 weeks	Advanced
Infrastructure as Code (Terraform)	3-4 weeks	Advanced
Monitoring and Logging (Prometheus)	3-4 weeks	Advanced
Total	10-14 months	